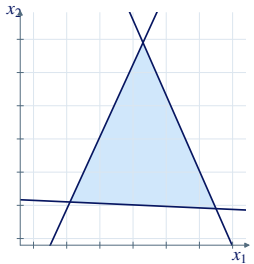
Proper Agnostic Learning of Functions of Halfspaces

Sergei Tikhonov (UT Austin) Arsen Vasilyan (Aarhus University)

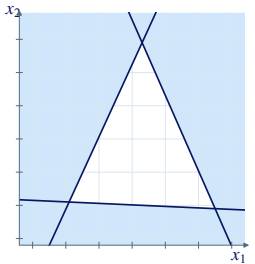
Boolean functions of halfspaces

A Boolean function of halfspaces is $B(h_1(x), \ldots, h_K(x)) \in \{\pm 1\}$, where each h_i is a halfspace.

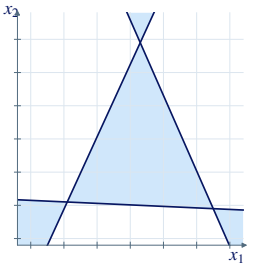
Intersection



Union



Parity (XOR)



and many others!

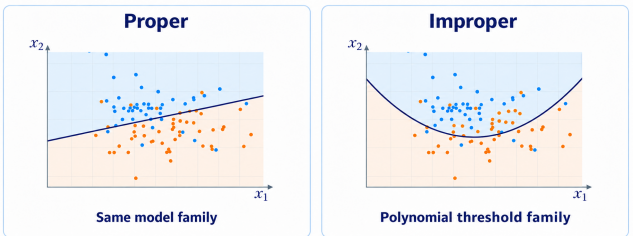
Proper agnostic learning

We study **distribution-specific agnostic learning**: covariates $x \sim \mathcal{N}(0, I_d)$, while the labels $y \in \{\pm 1\}$ are arbitrary.

Let $\mathcal{C}_K$ be the class of all Boolean functions of K halfspaces. **Our goal** is to learn $h \in \mathcal{C}_K$ satisfying, with high probability,

$$\Pr_{(x,y)\sim D}[h(x) \neq y] \leq \mathrm{OPT}_K + \varepsilon.$$

Proper learning returns a member of $\mathcal{C}_K$, while **improper learning** returns a function that is not necessarily in $\mathcal{C}_K$.



Why proper algorithm is better than improper:

	Proper learner	Improper learner
Stored parameters	$d+1$	$\binom{d+k}{k}$
Cost per prediction	$O(d)$	$O\left(\binom{d+k}{k}\right)$
Boundary volume	Stable	Unstable
Error guarantee	Best halfspace error $+\varepsilon$	Best halfspace error $+\varepsilon$

Boundary volume measures how often a perturbation can flip the label.

Our results

Proper learning matches the **dimension dependence** of improper learning.

Concept class	Our runtime	Best improper algorithm	SQ lower bound
Halfspaces	$d^{O(1/\varepsilon^2)} + S_H$	$d^{O(1/\varepsilon^2)}$	$d^{\Omega(1/\varepsilon^2)}$
Intersections of K halfspaces	$d^{O\left(\frac{\log K \, \log(1/\varepsilon)}{\varepsilon^2}\right)} + S_I$	$d^{O\left(\frac{\log K \, \log(1/\varepsilon)}{\varepsilon^2}\right)}$	$d^{\Omega\left(\frac{\log K}{\log\log K} + \frac{1}{\varepsilon^2}\right)}$
Boolean functions of K halfspaces	$d^{O\left(\frac{K^2 \log(1/\varepsilon)}{\varepsilon^2}\right)} + S_B$	$d^{O\left(\frac{K^2 \log(1/\varepsilon)}{\varepsilon^2}\right)}$	$d^{\Omega(K^2/\varepsilon^2)}$

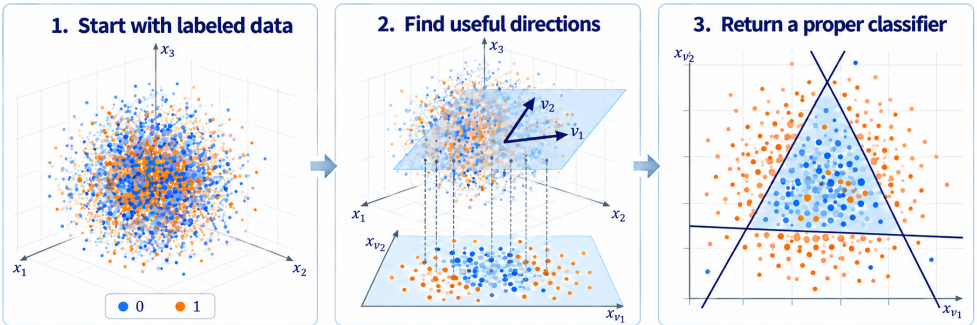
Additional search cost (independent of d):

$$S_H = (1/\varepsilon)^{O(1/\varepsilon^{5/2})} \qquad S_I = \left(\sqrt{\log K}/\varepsilon\right)^{O\left(K^2\sqrt{\log K}/\varepsilon^{5/2}\right)} \qquad S_B = (K/\varepsilon)^{O(K^3/\varepsilon^{5/2})}$$

Improper: Klivans et al. (2008); Pesenti et al. (2026).
SQ: Diakonikolas et al. (2021); Hsu et al. (2022).

Algorithmic overview

Idea: Use an improper classifier to find a low-dimensional subspace that supports a near-optimal proper classifier.



Algorithm: functions of K halfspaces

Goal: return a Boolean function of K halfspaces with error $\mathrm{OPT}_K + \varepsilon$.

Objective: $\mathcal{L}(P) = \underbrace{\mathbb{E}[\log(1 + e^{-yP(x)})]}_{\text{fit the labels}} + \underbrace{\mu\,\mathbb{E}[P(x)^2]}_{\text{ensure strong convexity}} + \underbrace{\nu\,\mathrm{tr}(M(P)^{1/2})}_{\text{control the search dimension}}$

1. **Learn P.** Fit a low-degree polynomial to the labels using regularized logistic loss.
2. **Construct V.** Keep the large eigenvalue directions of $M(P) = \mathbb{E}[\nabla P(x) \nabla P(x)^{\mathsf{T}}]$.
3. **Search within V.** Try joint choices of K normals from a finite cover and K thresholds from a grid.
4. **Select the best candidate.** On one fresh labeled sample, fit each candidate's Boolean rule by majority labels. Return the classifier with the fewest mistakes on this sample.

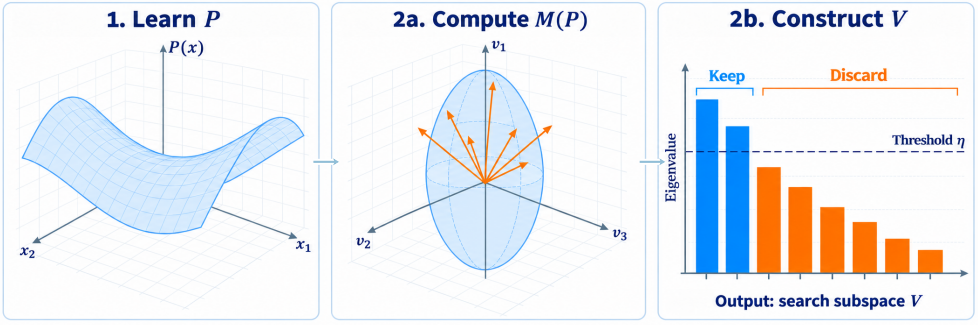
Constructing low-dimensional subspace V

1. Optimize the regularized objective:

$$\mathcal{L}(P) = \underbrace{\mathbb{E}[\log(1+e^{-yP(x)})]}_{\text{fit the labels}} + \underbrace{\mu\,\mathbb{E}[P(x)^2]}_{\text{ensure strong convexity}} + \underbrace{\nu\,\mathrm{tr}(M(P)^{1/2})}_{\text{control the search dimension}}$$

2a. Construct influence matrix: $M(P) = \mathbb{E}[\nabla P(x) \nabla P(x)^\mathsf{T}]$.

2b. Choose subspace: $V = \mathrm{span}\{v_i : \lambda_i \geq \eta\}$.



Searching within low-dimensional subspace V

3a. Search over candidate normals in V.

3b. Search over candidate thresholds.

3c. Fit the Boolean rule using the majority label in each cell.

